

QuantMS nextflow workflow (nf-core compatible)

23th February 2026

Henry Webel

Does broccoli boost bad gut bacteria?

- One strain of *E. coli* was analyzed using MS-based proteomics
- Shoutout to [Caroline Jachmann](#) for creating a peptide atlas for *E. coli* and sharing information on small scale experiments
- DDA dataset

<https://www.hudson.org.au/news/does-broccoli-boost-bad-gut-bacteria/>

INFLAMMATION

NEWS | INFLAMMATION | MICROBIOME IN HEALTH AND DISEASE

Does broccoli boost bad gut bacteria?

22 August 2023

By Rob Clancy, staff writer. Reviewed by Dr Emily Gulliver



Dr Emily Gulliver

Latest research into the human microbiome begins to untangle how broccoli can alter healthy gut bacteria.

Cruciferous vegetables, including broccoli, cauliflower, brussels sprouts and kale, are often recommended due to the presence of the antioxidant sulforaphane, which is thought to be beneficial for general health and wellbeing and in treating diseases such as cancer.

Mass spectrometry-based
Proteomics introduction
documentation



Search

% + K

QuantMS

QuantMS Hands-On for
PXD04621

QuantMS Input Files

Example datasets



QuantMS Hands-On for PXD04621

- [slides](#)

Open GitHub codespace

Use the following link to open a GitHub codespace with most of the required software installed:

⚠ If you do it manually, make sure to select the bigger machine with 4 cores and 16GB RAM



Open in GitHub Codespaces

1

2

Codespace with 4 cores (and 16GB of memory)

Create codespace for **biosustain/dsp_course_proteomics_intro**

Branch This branch will be checked out on creation	 main ▾
Dev container configuration Your codespace will use this configuration	nextflow-training ▾
Region Your codespace will run in the selected region	Europe West ▾
Machine type Resources for your codespace	4-core ▾
Create codespace	

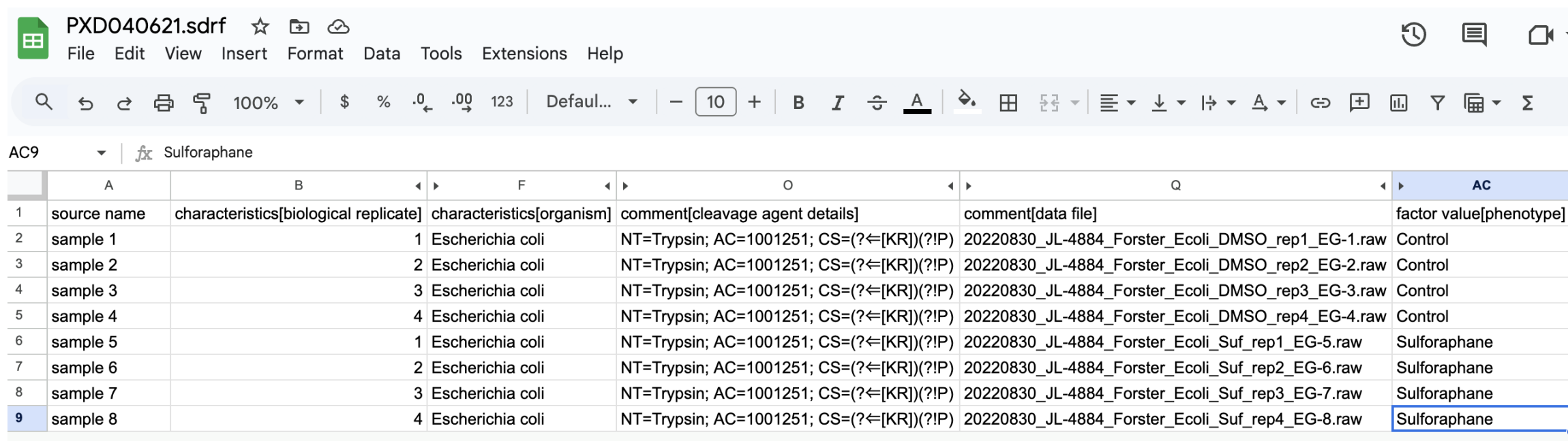
1

2

INPUTS

Sample Data Relationship Format (SDRF) file

Describes the design of an experiment (metadata): Defines in a standardized manner details about how the experiment was performed - sample information, organism, phenotype/grouping, etc.



	A	B	F	O	Q	AC
1	source name	characteristics[biological replicate]	characteristics[organism]	comment[cleavage agent details]	comment[data file]	factor value[phenotype]
2	sample 1		1 Escherichia coli	NT=Trypsin; AC=1001251; CS=?<[KR])(?!P)	20220830_JL-4884_Forster_Ecoli_DMSO_rep1_EG-1.raw	Control
3	sample 2		2 Escherichia coli	NT=Trypsin; AC=1001251; CS=?<[KR])(?!P)	20220830_JL-4884_Forster_Ecoli_DMSO_rep2_EG-2.raw	Control
4	sample 3		3 Escherichia coli	NT=Trypsin; AC=1001251; CS=?<[KR])(?!P)	20220830_JL-4884_Forster_Ecoli_DMSO_rep3_EG-3.raw	Control
5	sample 4		4 Escherichia coli	NT=Trypsin; AC=1001251; CS=?<[KR])(?!P)	20220830_JL-4884_Forster_Ecoli_DMSO_rep4_EG-4.raw	Control
6	sample 5		1 Escherichia coli	NT=Trypsin; AC=1001251; CS=?<[KR])(?!P)	20220830_JL-4884_Forster_Ecoli_Suf_rep1_EG-5.raw	Sulforaphane
7	sample 6		2 Escherichia coli	NT=Trypsin; AC=1001251; CS=?<[KR])(?!P)	20220830_JL-4884_Forster_Ecoli_Suf_rep2_EG-6.raw	Sulforaphane
8	sample 7		3 Escherichia coli	NT=Trypsin; AC=1001251; CS=?<[KR])(?!P)	20220830_JL-4884_Forster_Ecoli_Suf_rep3_EG-7.raw	Sulforaphane
9	sample 8		4 Escherichia coli	NT=Trypsin; AC=1001251; CS=?<[KR])(?!P)	20220830_JL-4884_Forster_Ecoli_Suf_rep4_EG-8.raw	Sulforaphane

LessSDRF
tool helps to
create these
tables

[View online](#)

...

>sp|PODSF9|EVGL_ECOLI Protein EvgL OS=Escherichia coli (strain K12) OX=83333 GN=evgL PE=1 SV=1

MLHCKGNNL

>sp|PODSH1|YSAE_ECOLI Protein YsaE OS=Escherichia coli (strain K12) OX=83333 GN=ysaE PE=1 SV=1

MRNAVSKAGIISRRRLLLFQFAG

>sp|PODV20|YTCB_ECOLI Protein YtcB OS=Escherichia coli (strain K12) OX=83333 GN=yticB PE=1 SV=1

MHLQLIKDNIHSVVICYT

...

>CON_ENSEMBL:ENSBTAP00000038329 (Bos taurus) 9 kDa protein

LPENVTPPEQHKGTSVIHKAVLDVGEEGTEGA AVTAVVMATSSLLHTLVSFNRPFLLSI

FCKETQSIIFLGKVTNPKEA

...

- 4413 known *E. coli* proteins
- 246 known contaminants sequences

- ThermoFisher instruments:
 - raw files
- Open standard, text based:
 - mzML files
- In this tutorial we use mzML files directly to skip the spectra extraction as quantms run on mzML files

One MS1 spectrum (DDA)

```
<spectrum id="controllerType=0 controllerNumber=1 scan=6" index="5" defaultArrayLength="736">
  <cvParam cvRef="MS" accession="MS:1000511" value="1" name="ms level" />
  <cvParam cvRef="MS" accession="MS:1000579" value="" name="MS1 spectrum" />
  <cvParam cvRef="MS" accession="MS:1000130" value="" name="positive scan" />
  <cvParam cvRef="MS" accession="MS:1000285" value="15248891" name="total ion current" />
  <cvParam cvRef="MS" accession="MS:1000127" value="" name="centroid spectrum" />
  <cvParam cvRef="MS" accession="MS:1000504" value="401.923461914063" name="base peak m/z" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
  <cvParam cvRef="MS" accession="MS:1000505" value="2704116.25" name="base peak intensity" unitAccession="MS:1000131" unitName="number of detector counts" unitCvRef="MS" />
  <cvParam cvRef="MS" accession="MS:1000528" value="375.884124755859" name="lowest observed m/z" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
  <cvParam cvRef="MS" accession="MS:1000527" value="1665.40612792969" name="highest observed m/z" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
  <scanList count="1">
    <cvParam cvRef="MS" accession="MS:1000795" value="" name="no combination" />
    <scan instrumentConfigurationRef="IC1">
      <cvParam cvRef="MS" accession="MS:1000016" value="6.0323342" name="scan start time" unitAccession="U0:0000031" unitName="minute" unitCvRef="U0" />
      <cvParam cvRef="MS" accession="MS:1000512" value="FTMS + p NSI Full ms [375.0000-1800.0000]" name="filter string" />
      <cvParam cvRef="MS" accession="MS:1000927" value="50" name="ion injection time" unitAccession="U0:0000028" unitName="millisecond" unitCvRef="U0" />
      <scanWindowList count="1">
        <scanWindow>
          <cvParam cvRef="MS" accession="MS:1000501" value="375" name="scan window lower limit" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
          <cvParam cvRef="MS" accession="MS:1000500" value="1800" name="scan window upper limit" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
        </scanWindow>
      </scanWindowList>
    </scan>
  </scanList>
  <binaryDataArrayList count="2">
    <binaryDataArray encodedLength="3152">
      <cvParam cvRef="MS" accession="MS:1000514" value="" name="m/z array" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
      <cvParam cvRef="MS" accession="MS:1000523" value="" name="64-bit float" />
      <cvParam cvRef="MS" accession="MS:1000574" value="" name="zlib compression" />
      <binary>eJwT03LYVXuawPHjVumjGA1lG5dTGtMi3MYUG32496hJOiNX3EvLcJLNsEDNckGWlYogXaIwiU3AsUWcZ2TJEnEm5Wimg48C466j15M12ZRgU1paCTPP+/399Xne9fceFk3TzIeylxna/0Vjck5ovM26h1oDLdI
    </binaryDataArray>
    <binaryDataArray encodedLength="4356">
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      <cvParam cvRef="MS" accession="MS:1000523" value="" name="64-bit float" />
      <cvParam cvRef="MS" accession="MS:1000574" value="" name="zlib compression" />
      <binary>eJwTWHlct+kXvILFSI2MTJi5ZWlATGdSl+Ekt3KVtypUVl+dhr5TG4qo7Jk8FPWm32pjC1kuxULeu0JuSVDpozK0pdKpj3P/PV+3vu+73nPec45zznvFQRBUrdyQxIEQWzvf6ppFBz63sIYceZq06jafJHbNGr
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  </binaryDataArrayList>
</spectrum>
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  <cvParam cvRef="MS" accession="MS:1000505" value="2704116.25" name="base peak intensity" unitAccession="MS:1000131" unitName="number of detector counts" unitCvRef="MS" />
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  <cvParam cvRef="MS" accession="MS:1000527" value="1665.40612792969" name="highest observed m/z" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
  <scanList count="1">
    <cvParam cvRef="MS" accession="MS:1000795" value="" name="no combination" />
    <scan instrumentConfigurationRef="IC1">
      <cvParam cvRef="MS" accession="MS:1000016" value="6.0323342" name="scan start time" unitAccession="UO:0000031" unitName="minute" unitCvRef="UO" />
      <cvParam cvRef="MS" accession="MS:1000512" value="FTMS + p NSI Full ms [375.0000-1800.0000]" name="filter string" />
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    </scan>
  </scanList>
  <scanWindowList count="1">
    <scanWindow>
      <cvParam cvRef="MS" accession="MS:1000501" value="375" name="scan window lower limit" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
      <cvParam cvRef="MS" accession="MS:1000500" value="1800" name="scan window upper limit" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
    </scanWindow>
  </scanWindowList>
</spectrum>
<binaryDataArray encodedLength="3152">
  <cvParam cvRef="MS" accession="MS:1000514" value="" name="m/z array" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
  <cvParam cvRef="MS" accession="MS:1000523" value="" name="64-bit float" />
  <cvParam cvRef="MS" accession="MS:1000574" value="" name="zlib compression" />
  <binary> ... 93jb+B7DdOsY=</binary>
</binaryDataArray>
<binaryDataArray encodedLength="4356">
  <cvParam cvRef="MS" accession="MS:1000515" value="" name="intensity array" unitAccession="MS:1000131" unitName="number of counts" unitCvRef="MS" />
  <cvParam cvRef="MS" accession="MS:1000523" value="" name="64-bit float" />
  <cvParam cvRef="MS" accession="MS:1000574" value="" name="zlib compression" />
  <binary>eJwTWHlCt+ ... yNfr3NnHXsNq2F/69YyX/gWrVuzh</binary>
</binaryDataArray>
</binaryDataArrayList>
</spectrum>
```

QuantMS (DDA)

- Nextflow is a workflow executor (analysis steps combined in an execution graph)
 - Reproducible and scalable
- Nf-core is a collection of workflows maintained by nextflow and an open-source community



A global community effort to collect a curated set of open-source analysis pipelines built using Nextflow.

Pipelines

Browse the 130 pipelines that are currently available as part of nf-core.

Released 80 Under development 38 Archived 12

↓ Last release ▾



proteinfamilies ✓

☆ 14

Generation and update of protein families

metagenomics protein-families proteomics

1.1.0 released 4 days ago

epitopeprediction ✓

☆ 44

A bioinformatics best-practice analysis pipeline for epitope prediction and annotation

epitope epitope-prediction mhc-binding-prediction

3.0.0 released 5 days ago

Works with

- Linux machines (on servers)
- on MacOS

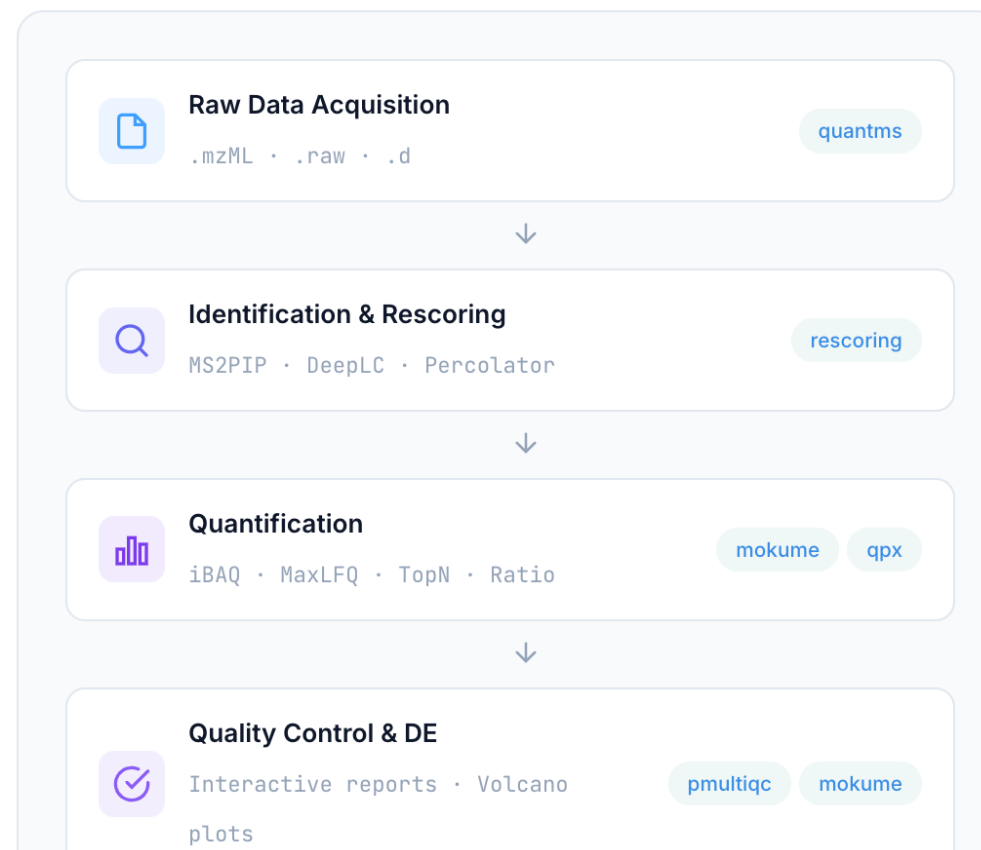
A Nextflow-orchestrated workflow where specialized tools handle each step.

End-to-End Quantitative Proteomics

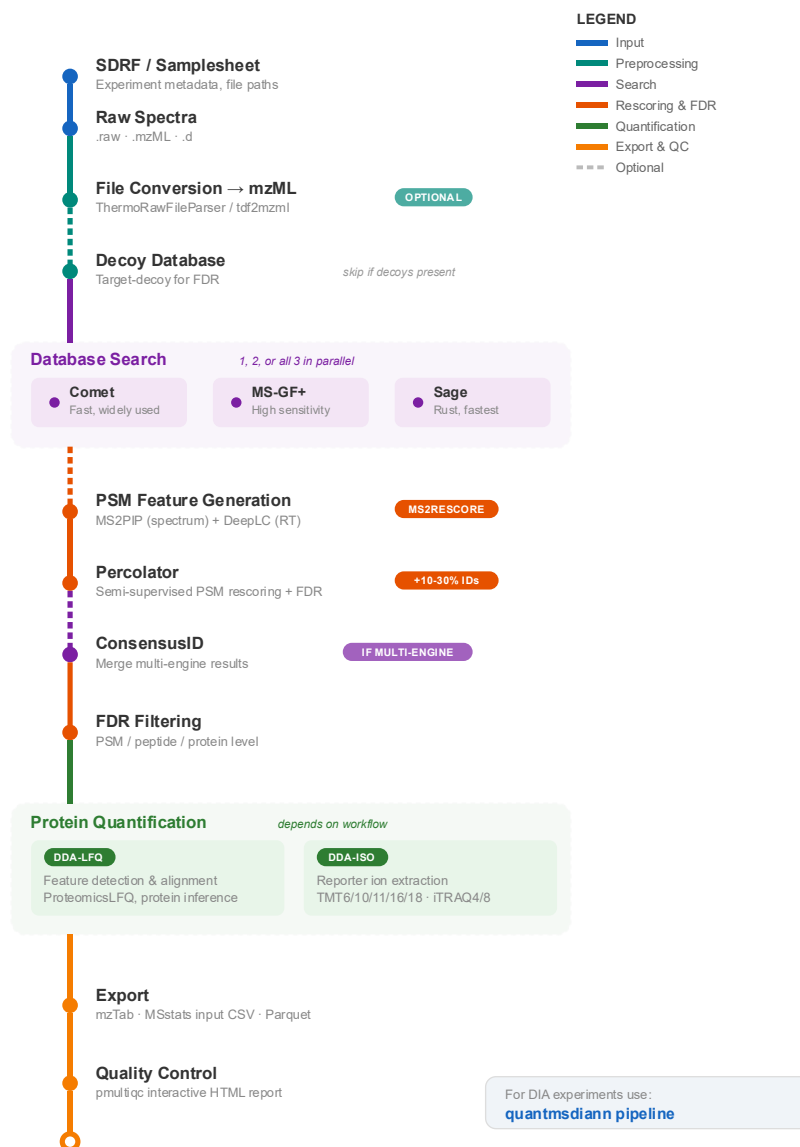
Two dedicated Nextflow pipelines: **quantms** for DDA (LFQ & TMT/iTRAQ) and **quantmsdiann** for DIA — but the downstream tools work independently too. Use **mokume** to quantify proteins from any search engine output. Use **pmultiqc** for QC on any experiment.

- 1 **Convert & search** raw spectra against protein databases
- 2 **Rescore PSMs** with ML models for higher sensitivity
- 3 **Quantify proteins** using iBAQ, MaxLFQ, TopN, or ratio methods
- 4 **Generate QC reports** and differential expression results

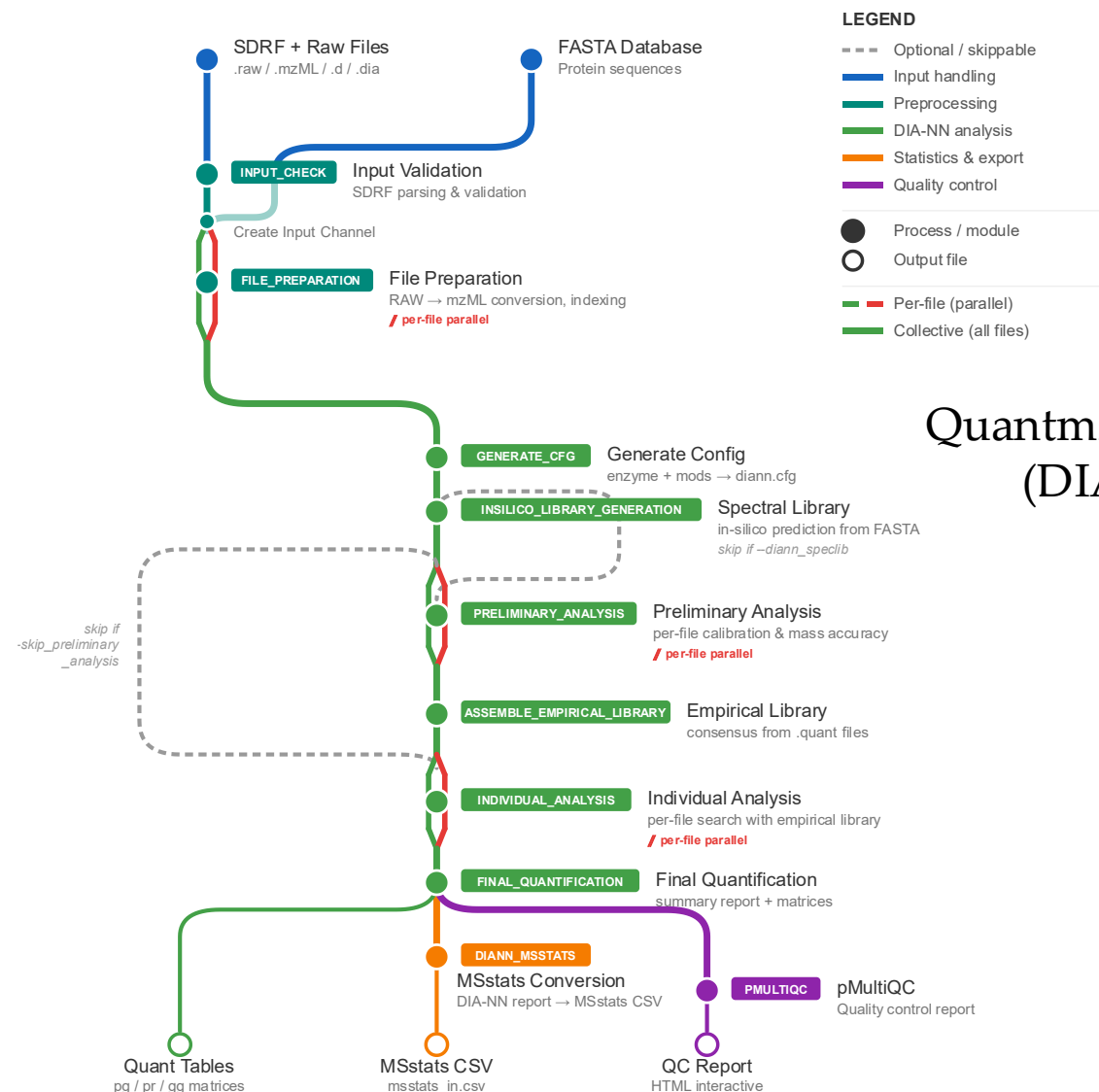
[Read pipeline docs →](#)



QuantMS (DDA)

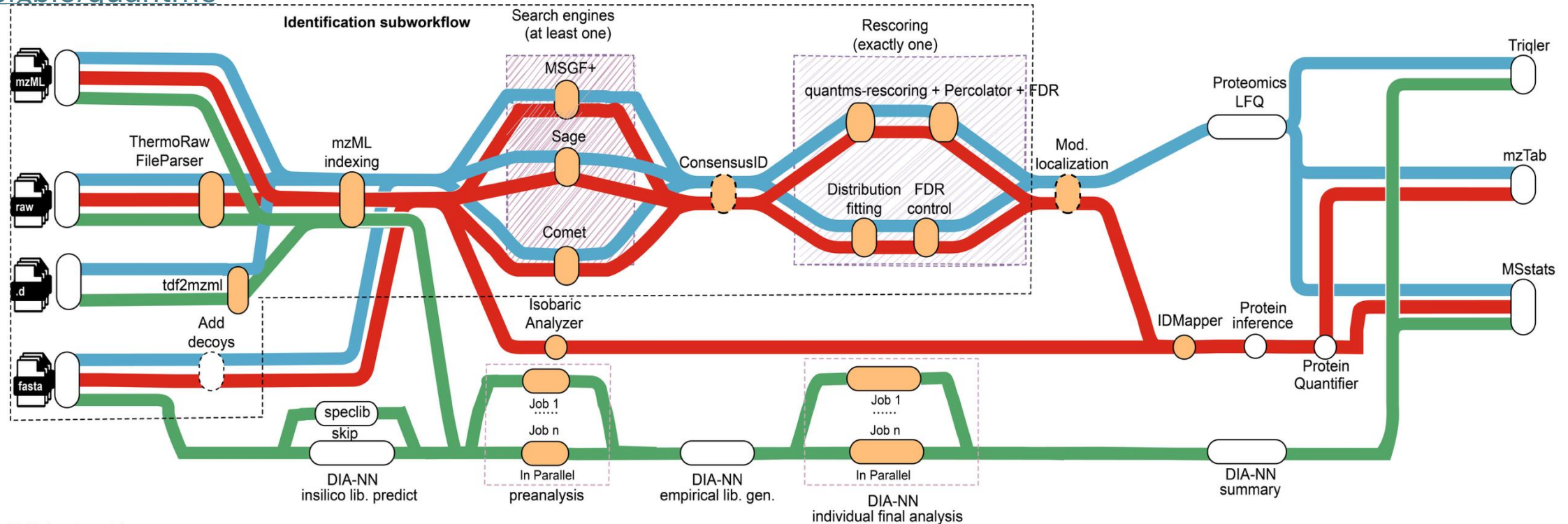


Quantmsdiann (DIA)



QuantMS (nf-core compatible) v1.7.0 and earlier

bigbio/quantms



bigbio/quantms v1.3

Example analysis pathways

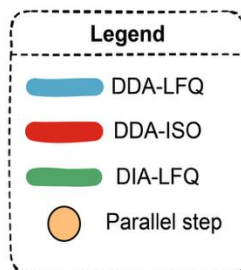


Table 8 Capabilities timeline

Capability	Introduced in	Notes
DDA-LFQ workflow	1.0.0	Label-free quantification; OpenMS-based quantification. See Release 1.0.0 .
Isobaric TMT workflow	1.2.0	TMT reporter quantification and MSstats(TMT) export. See Release 1.2.0 .
DIA workflow (DIA-NN based)	1.3.0	DIA identification/quantification and exports to MSstats and Triqler. See Release 1.3.0 .
Conda disabled by default	1.4.0	Container-first execution (Docker/Singularity/Podman). See Release 1.4.0 .
quantms-rescoring (DeepLC/MS2PIP integration)	1.5.0	Optional ML-based rescoring features for improved identifications. See Release 1.5.0 .
1.6.0 highlights	1.6.0	AlphaPeptDeep integration into quantms-rescoring. See Release 1.6.0 .
1.7.0	1.7.0	quantms-rescoring supports AlphaPeptDeep transfer learning. LuciPHORAdapter is replaced with onsite python package. See Release 1.7.0 .

See overview [here](#)

- Cloud native
- Containerized
- Focused on reproducibility and scalability
- Integrates OpenMS and DIA-NN
- Integrates three search engines: MSGF+, Sage and Comet (which can be all run for a search)

results/PXD040621

pipeline_info # parameters etc

pmultiqc # MultiQC report with file level quant information

quant_tables # peptide-level quantification

sdrf # Sample-Data-Relationship-Format (parsed and validated)

spectra # mzML files indexed (if raw files were parsed)

database: data/fasta/merged_ecoli_with_contaminants.fasta

input: data/PXD040621/PXD040621.sdrf.tsv

outdir: results/PXD040621

only relevant for 1.3.0

max_memory: 15 GB

max_cpus: 4

local_input_type: raw

local_input_type: mzML # default

! root_folder only applies to the mzML or raw spectrum files

root_folder: /workspaces/dsp_course_proteomics_intro/data/PXD040621/mzML/

publish_dir_mode: symlink

running msstats: Only two pdfs which we do not use.

skip_post_msstats: true

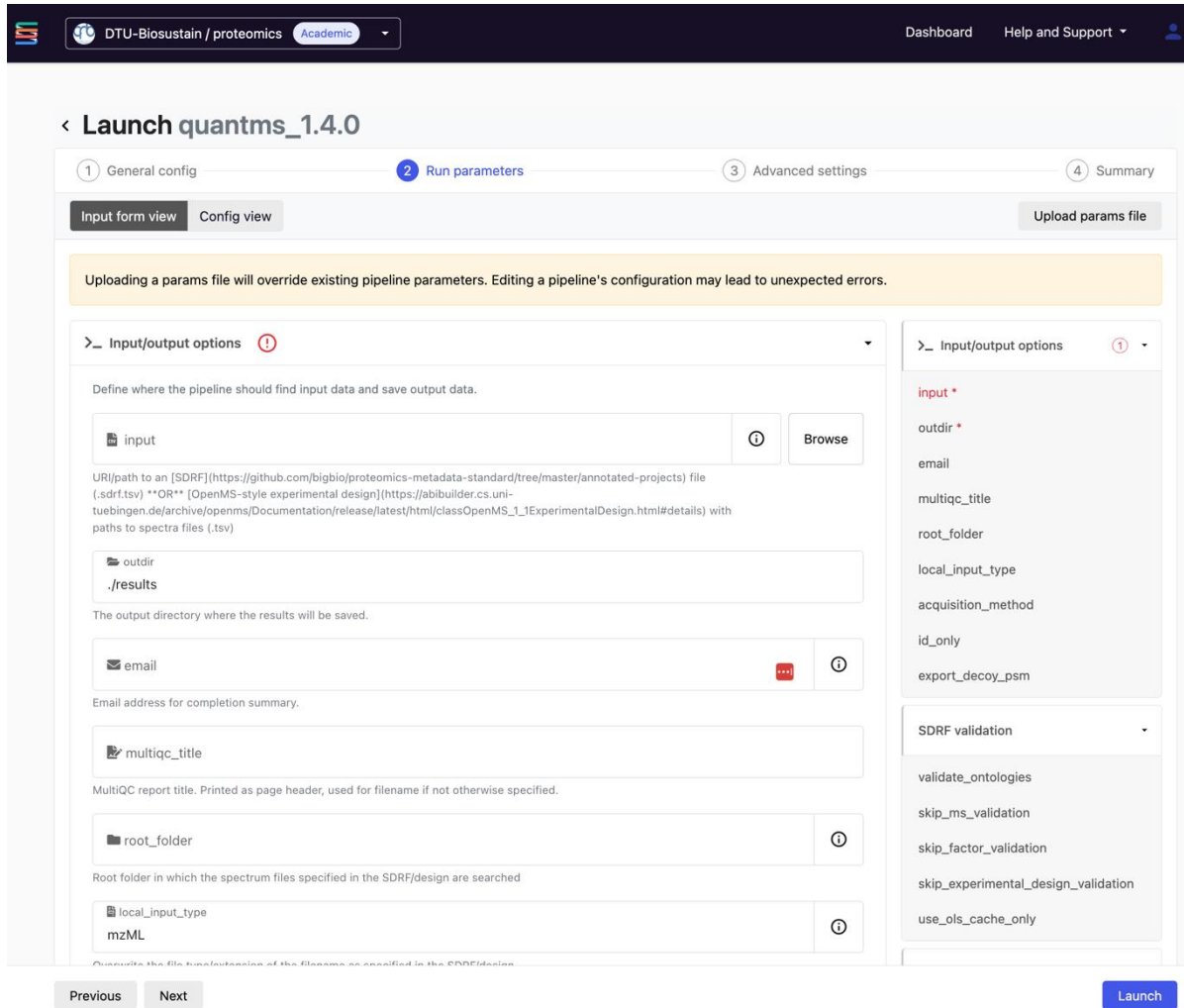
min_peptide_length: 6
max_peptide_length: 40

fdr_level: psm_level_fdrs
min_precursor_charge: 2
max_precursor_charge: 4

protein_quant: unique_peptides
min_peptides_per_protein: 1

precursor_mass_tolerance: 5
precursor_mass_tolerance_unit: ppm
fragment_mass_tolerance: 0.03
fragment_mass_tolerance_unit: Da
protein_level_fdr_cutoff: 0.01

Seqera interface for workflow



DTU-Biosustain / proteomics Academic

Dashboard Help and Support

< Launch quantms_1.4.0

1 General config 2 Run parameters 3 Advanced settings 4 Summary

Input form view Config view Upload params file

Uploading a params file will override existing pipeline parameters. Editing a pipeline's configuration may lead to unexpected errors.

> Input/output options ⓘ

Define where the pipeline should find input data and save output data.

input ⓘ Browse

URI/path to an [SDRF](https://github.com/bigbio/proteomics-metadata-standard/tree/master/annotated-projects) file (.sdrf.tsv) **OR** [OpenMS-style experimental design](https://abibuilder.cs.uni-tuebingen.de/archive/openms/Documentation/release/latest/html/classOpenMS_1_1ExperimentalDesign.html#details) with paths to spectra files (.tsv)

outdir
./results

The output directory where the results will be saved.

email ⓘ

Email address for completion summary.

multiqc_title

MultiQC report title. Printed as page header, used for filename if not otherwise specified.

root_folder ⓘ

Root folder in which the spectrum files specified in the SDRF/design are searched

local_input_type ⓘ

mzML

Configure the file type/extension of the filename as specified in the SDRF/design

input *
outdir *
email
multiqc_title
root_folder
local_input_type
acquisition_method
id_only
export_decoy_psm

SDRF validation

validate_ontologies
skip_ms_validation
skip_factor_validation
skip_experimental_design_validation
use_ols_cache_only

Previous Next Launch

- Parameters grouped and in order
- Based on [schema file](#)
 - Schema file can be launched to see it website locally
- For more information: [Schema file documentation](#)
- Remember: Show live in a workspace

Inspect some of the steps

1. SAMPLESHEET_CHECK
2. SDRF_PARSING
3. MZML_INDEXING
4. MZML_STATISTICS
5. GENERATE_DECOY_DATABASE
6. PEPTIDE_DATABASE_SEARCH:Comet
7. PERCOLATOR
8. ID_FILTER
9. PROTEOMICS_LFQ

```
NEXTFLOW ~ version 25.10.4
Launching 'https://github.com/bigbio/quantms' [shrivelled_nobel] DSL2 - revision: e719f43a9f [1.7.0]

executor > local (1)
[8f/48b7ed] process > BIGBIO_QUANTMS:QUANTMS:INPUT_CHECK:SAMPLESHEET_CHECK (PXD040621.sdrf.tsv) [100%] 1 of 1, cached: 1 ✓
[82/a60364] process > BIGBIO_QUANTMS:QUANTMS:CREATE_INPUT_CHANNEL:SDRF_PARSING (PXD040621.sdrf.tsv) [100%] 1 of 1, cached: 1 ✓
[-] process > BIGBIO_QUANTMS:QUANTMS:FILE_PREPARATION:DECOMPRESS -
[09/fdb785] process > BIGBIO_QUANTMS:QUANTMS:FILE_PREPARATION:MZML_INDEXING (20220830_JL-4884_Forster_Ecoli_Suf_rep4_EG-8) [100%] 8 of 8, cached: 8 ✓
[-] process > BIGBIO_QUANTMS:QUANTMS:FILE_PREPARATION:THERMORAWFILEPARSER -
[d0/21ec6d] process > BIGBIO_QUANTMS:QUANTMS:FILE_PREPARATION:MZML_STATISTICS (20220830_JL-4884_Forster_Ecoli_Suf_rep4_EG-8) [100%] 8 of 8, cached: 8 ✓
[0f/cb7fbb] process > BIGBIO_QUANTMS:QUANTMS:GENERATE_DECOY_DATABASE (1) [100%] 1 of 1, cached: 1 ✓
[-] process > BIGBIO_QUANTMS:QUANTMS:TMT:ID:PEPTIDE_DATABASE_SEARCH:COMET -
[-] process > BIGBIO_QUANTMS:QUANTMS:TMT:ID:PSM_RESCORING:PERCOLATOR -
[-] process > BIGBIO_QUANTMS:QUANTMS:TMT:ID:PSM_FDR_CONTROL:ID_FILTER -
[-] process > BIGBIO_QUANTMS:QUANTMS:TMT:FEATURE_MAPPER:ISOBARIC_ANALYZER -
[-] process > BIGBIO_QUANTMS:QUANTMS:TMT:FEATURE_MAPPER:ID_MAPPER -
[-] process > BIGBIO_QUANTMS:QUANTMS:TMT:FILE_MERGE -
[-] process > BIGBIO_QUANTMS:QUANTMS:TMT:PROTEIN_INFERENCE:PROTEIN_INFERENCE_GENERIC -
[-] process > BIGBIO_QUANTMS:QUANTMS:TMT:PROTEIN_INFERENCE:ID_FILTER -
[-] process > BIGBIO_QUANTMS:QUANTMS:TMT:PROTEIN_QUANT:ID_CONFLICT_RESOLVER -
[-] process > BIGBIO_QUANTMS:QUANTMS:TMT:PROTEIN_QUANT:PROTEIN_QUANTIFIER -
[-] process > BIGBIO_QUANTMS:QUANTMS:TMT:PROTEIN_QUANT:MSSTATS_CONVERTER -
[af/641be9] process > BIGBIO_QUANTMS:QUANTMS:LFQ:ID:PEPTIDE_DATABASE_SEARCH:COMET (20220830_JL-4884_Forster_Ecoli_Suf_rep4_EG-8) [100%] 8 of 8, cached: 8 ✓
[bf/0377c0] process > BIGBIO_QUANTMS:QUANTMS:LFQ:ID:PSM_RESCORING:PERCOLATOR (20220830_JL-4884_Forster_Ecoli_Suf_rep4_EG-8) [100%] 8 of 8, cached: 8 ✓
[6a/40734d] process > BIGBIO_QUANTMS:QUANTMS:LFQ:ID:PSM_FDR_CONTROL:ID_FILTER (20220830_JL-4884_Forster_Ecoli_Suf_rep4_EG-8) [100%] 8 of 8, cached: 8 ✓
[28/717945] process > BIGBIO_QUANTMS:QUANTMS:LFQ:PROTEOMICS_LFQ (PXD040621.sdrf_openms_design) [100%] 1 of 1, cached: 1 ✓
[-] process > BIGBIO_QUANTMS:QUANTMS:DIA:GENERATE_CFG -
[-] process > BIGBIO_QUANTMS:QUANTMS:DIA:INSILICO_LIBRARY_GENERATION -
[-] process > BIGBIO_QUANTMS:QUANTMS:DIA:PRELIMINARY_ANALYSIS -
[-] process > BIGBIO_QUANTMS:QUANTMS:DIA:ASSEMBLE_EMPIRICAL_LIBRARY -
[-] process > BIGBIO_QUANTMS:QUANTMS:DIA:INDIVIDUAL_ANALYSIS -
[-] process > BIGBIO_QUANTMS:QUANTMS:DIA:FINAL_QUANTIFICATION -
[-] process > BIGBIO_QUANTMS:QUANTMS:DIA:CONVERT_RESULTS -
[f1/cb6a13] process > BIGBIO_QUANTMS:QUANTMS:SUMMARY_PIPELINE [ 0%] 0 of 1
```

Which ones run in parallel?

Example for data on website ([link](#))



Sections:

- Experimental Design and Metadata
- Results Overview
- Identification Summary
- Search Engine Scores
- Quantification Analysis
- MS1 Analysis
- MS2 and Spectral Stats
- Mass Error Trends
- RT Quality Control
- Software Versions



Technological Innovation and Resources

pmultiqc: An Open-Source, Lightweight, and Metadata-Oriented QC Reporting Library for MS Proteomics

Qi-Xuan Yue¹, Chengxin Dai², Selvakumar Kamatchinathan³, Chakradhar Bandla³, Henry Weibel⁴, Asier Larrea^{3,5}, Wout Bittremieux⁶, Julian Uszkoreit^{7,8}, Tom David Müller^{9,10}, Jinqiu Xiao¹¹, Juergen Cox¹¹, Fengchao Yu¹², Philip Ewels¹³, Vadim Demichev¹⁴, Oliver Kohlbacher^{9,10,15}, Timo Sachsenberg^{9,10}, Chris Bielow¹⁶, Mingze Bai¹  , Yasset Perez-Riverol³  

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<https://doi.org/10.1016/j.mcpro.2026.101530>

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 Open access

Highlights

- Generate metadata-aware QC reports using the SDRF format.
- Interactive reports for DIA-NN, FragPipe, MaxQuant/MaxDIA, quantms workflows.
- Supports DIA-LFQ, DDA-LFQ, and plexDDA proteomics analyses.
- Open-source library extendable to new metrics, workflows, and formats.
- Public service to generate QC reports for datasets in the PRIDE database.

- MultiQC view in Seqera
- What to look out for
 - Do all samples roughly have the same number of quantified peptides?
 - (tbc)

Runs > Details

 **pensive_gutenberg**  

2xBtcQnJTy0WsX  quantms_1-6-0  bigbio/quantms  ...cratch/30NWx3cYA8Bm6n 

proteomics-comp-env  2026-06-02 09:16:14 

mzML  normalize  v170 

100% Workflow run progress

Tasks Logs Metrics Configuration Inputs **Outputs** Containers Run Info

Reports

Search report by name

Name	Kind	Size	Path	Actions
 MultiQC HTML report	html	3.18 MB	az://seqera/proteomics_course/PX...	

Quantmsdiann

- Run DIANN on a distributed system (with many cores)
- Not 100% like the Graphical User Interface (e.g. w.r.t to contaminants)
- Check-out dataset

Some BRIGHT (NNF Center for biotechnology research institute for the green transition) specifics

- Use DIA workflow on a Bruker TimsTof Pro 2
- Process the raw data using DIANN GUI
- We show here how to do it via Seqera via the Data Catalog
- Data Analysis based on DIANN output files


DIA-NN
05/28/2026 07:11:11
×
+

Experiment name: 05/28/2026 07:11:11

+

×

C:\Users\kzl465\OneDrive - Danmarks Tekniske Universitet\ms_datasets\PXD070899\raw\SB_4_64_SH_AY_R1.raw
C:\Users\kzl465\OneDrive - Danmarks Tekniske Universitet\ms_datasets\PXD070899\raw\SB_4_64_SH_AY_R2.raw
C:\Users\kzl465\OneDrive - Danmarks Tekniske Universitet\ms_datasets\PXD070899\raw\SB_4_64_SH_AY_R3.raw
C:\Users\kzl465\OneDrive - Danmarks Tekniske Universitet\ms_datasets\PXD070899\raw\SB_4_64_SH_AY18_R1.raw
C:\Users\kzl465\OneDrive - Danmarks Tekniske Universitet\ms_datasets\PXD070899\raw\SB_4_64_SH_AY18_R2.raw

Spectral library
C:\Users\kzl465\OneDrive - Danmarks Tekniske Unive ...

Calibration library
(optional) ...

C:\Users\kzl465\OneDrive - Danmarks Tekniske Universitet\ms_datasets\fasta_files\Saccharomyces_cerevisiae_UP2311_2025_11_23.fasta

☒ Contaminants

☐ Reannotate

Binary
diann.exe ...

Precursor ion generation

Mode
Prediction from library

Protease
Trypsin/P

☐ Semi

Misses
1

Var mods
0

+ Add/search modification...

N-term M excision × Carbamidomethyl (C) ×

Peptide length
7 - 30

Precursor m/z
300 - 1800

Precursor charge
1 - 4

Fragment m/z
200 - 1800

☐ Reuse .quant files

☒ Matrices

☐ XICs

☒ PDF

Main output
report.parquet ...

Temp / .dia directory
(optional) ...

Output library
lib.parquet ...

Predicted library
(filled automatically)

Additional options
--option value

☐ Detailed log

FDR %
5

Threads
12

Run

Stop

Algorithm

MS1 accuracy
0

MS2 accuracy
0

Cal accuracy
0

Scan window
0

☒ MBR

☒ Protein inference

☐ Unrelated runs

☐ Knowledge base

Scoring
Generic

Machine learning
NNs (cross-validated)

Cross-run normalisation
RT-dependent

Speed: peak filtering
Optimal results

Proteotypicity
Genes

Quantification strategy
QuantUMS (precision)

Library generation
IDs, RT & IM profiling

Speed: RT/IM filtering
Balanced

Gen license

About

*

Skyline

Analyse

Files

+

\$

×

^

v

Step name

↑

↓

Clear

--regex pattern replacement

Apply

Execute Pipeline

Abort

●

🔍

Search...

^

v

Save

Clear

- See example for DIANN based search using [PXD070899](#):
Capture and functional conversion of mitochondrial protein translocase by lipid metabolism enzyme
- Paper (<http://nature.com/articles/s41556-026-01890-3>):

Heinen, S. *et al.* Mitochondria contact lipid droplets through the mitochondrial import complex binding to lipid metabolism enzyme Ayr1. *Nat. Cell Biol.* **28**, 436–448 (2026).

Data Analysis hands-on using acore

14th May 2025

Henry Webel

- [illegible]

Aim: identify the impact of sulforaphane on the human gut microbiome (in anaerobic conditions of the gut)

Scientific Story:

- Phylogeny of selected strains and screening of growth kinetics of selected strains under sulforaphane in anaerobic conditions (Figure 1)
- Growth profile of selected *E. coli* Strain E2348/69 in anaerobic and aerobic conditions with varying amounts of sulforaphane
- Proteomics analysis of most growing strain
- Metabolomics analysis of most growing strain
- [Article](#) and [PXD040621](#) on Pride archive

“Proteomics identified an increase in anaerobic respiration in *E. coli* grown in the presence of sulforaphane, indicating suggesting that sulforaphane may be acting as an additional carbon source in these bacteria. The metabolic profile following growth in sulforaphane, showed that sulforaphane increased the production of metabolites that are also known to interact with host tissues to decrease inflammation.”

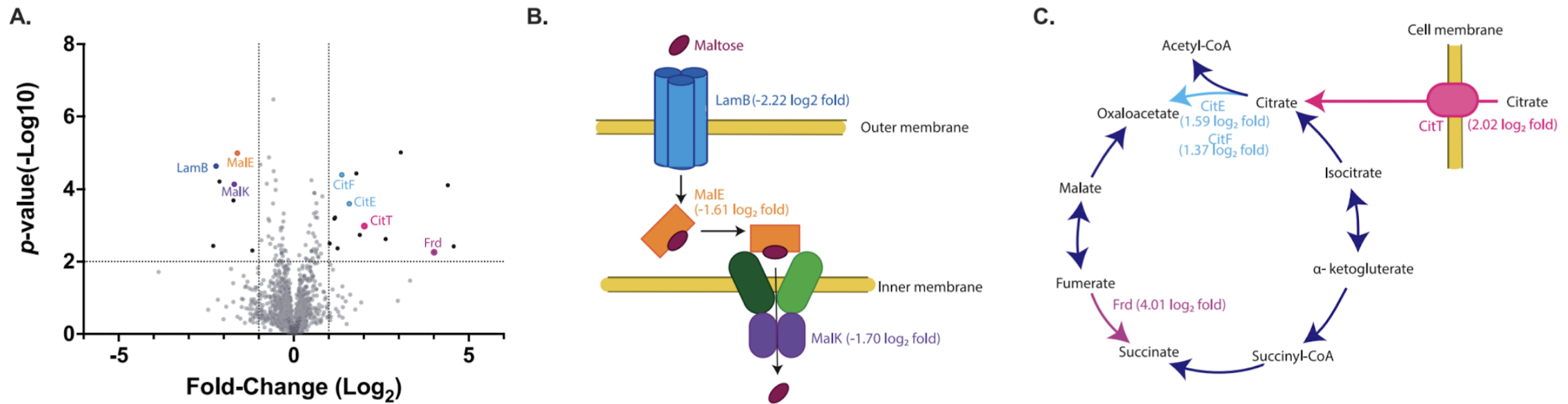


Fig. 3. Differentially produced proteins in *E. coli* EPEC ECE2348/69. **A.** Volcano plot of the change in protein production of individual proteins (dots) produced by *E. coli* ECE2348/69 in response to the presence of sulforaphane. Significantly altered proteins (black or coloured dots) showed a $\log_2$ fold-change < -1 or > 1 (vertical dashed lines) with a p -value < 0.05 (horizontal dashed lines). **B.** Schematic representation of proteins with significantly decreased production involved in maltose uptake. **C.** Schematic representation of proteins with significantly increased production (coloured labels) involved in anaerobic respiration.

MoNA and DSP Open Source Libraries



Analytics

Analytics core
library

analytics-core.readthedocs.io/



Visualisation



Visualization core
library

github.com/Multiomics-Analytics-Group/vuecore

Reporting

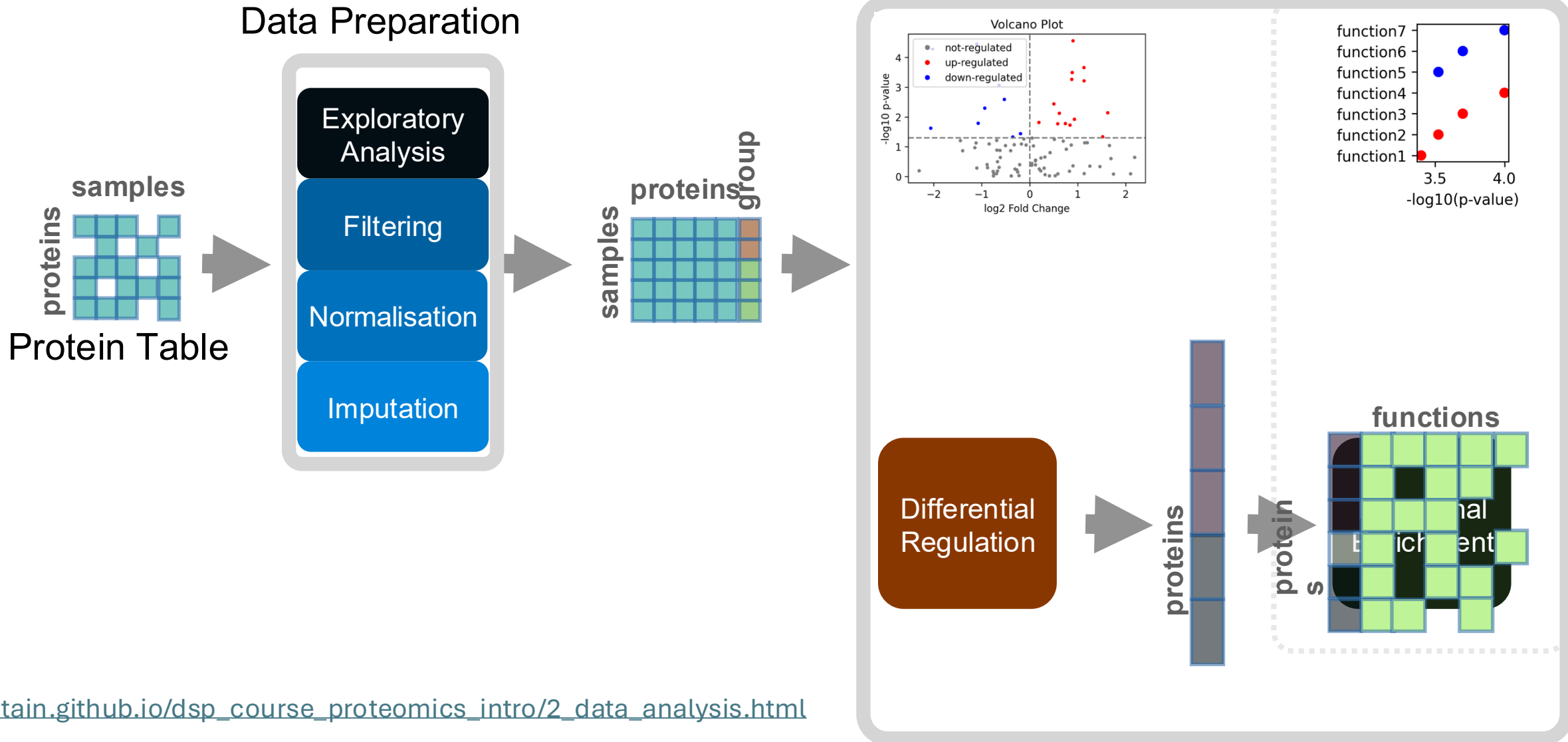


Automated Reporting
library and cli

github.com/Multiomics-Analytics-Group/vuegen



Acore – Analytical core – workflow example



- See rendered version on the website
- Focus on
 - Filtering
 - Normalization
 - Differential analysis (t-test with multiple testing correction)
 - Enrichment analysis
 - Comparison to original results

We discuss the outputs together, then you have time to experiment.

Mass spectrometry-based
Proteomics introduction
documentation



Search

% + K

QuantMS

QuantMS Hands-On for
PXD04621

QuantMS Input Files

Example datasets

Data Analysis

Data Analysis Hands-On for
PXD04621

[Data Analysis PXD040621](#)



Data Analysis PXD040621

Plan

- read data and log2 transform intensity values
- aggregate peptide intensities to protein intensities
- format data from long to wide format
- remove contaminant proteins
- check for missing values
- Clustermap of sample and proteins
- differential analysis (Volcano Plots)
- Enrichment Analysis
- check for maltose update pathway (Fig. 3 in paper)

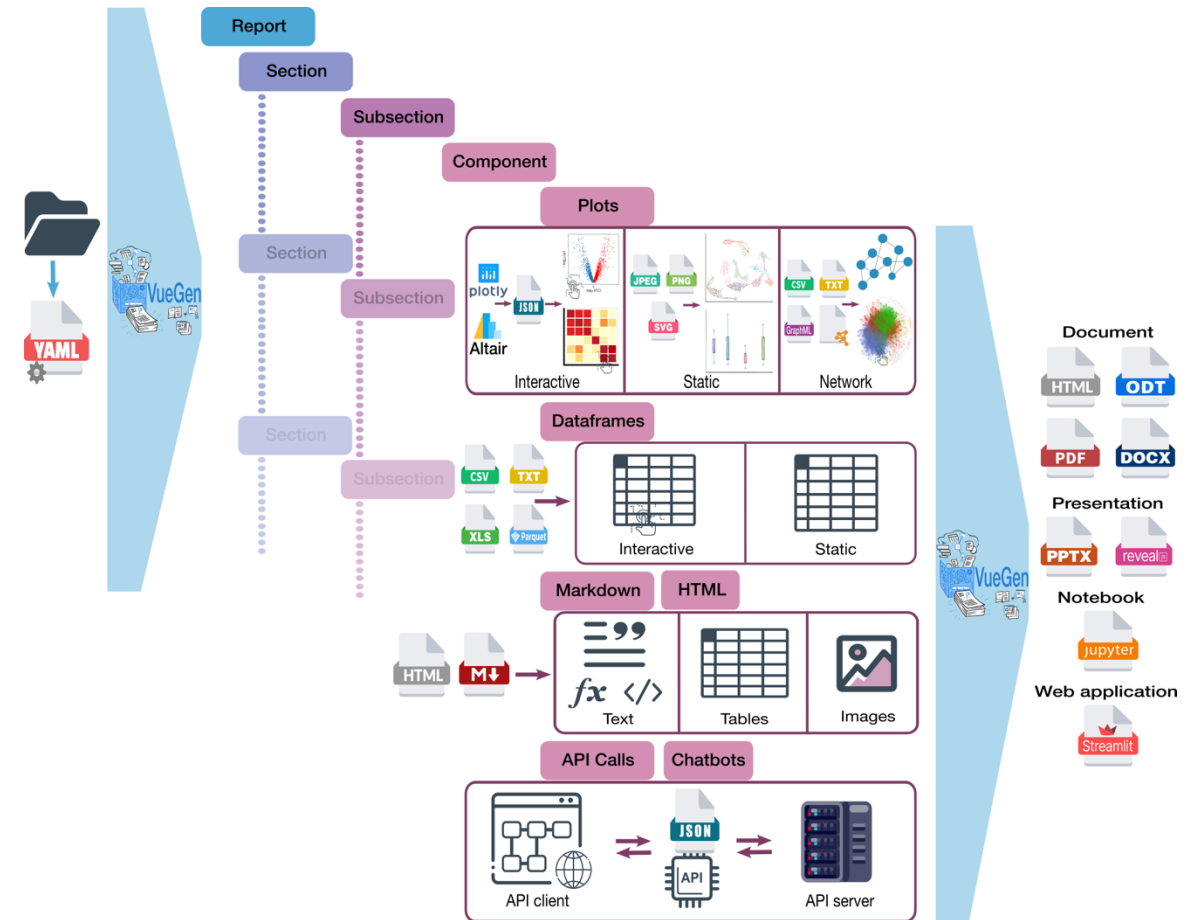
```
%pip install acore vuecore "pingouin<0.6.0"
```

► Show code cell output

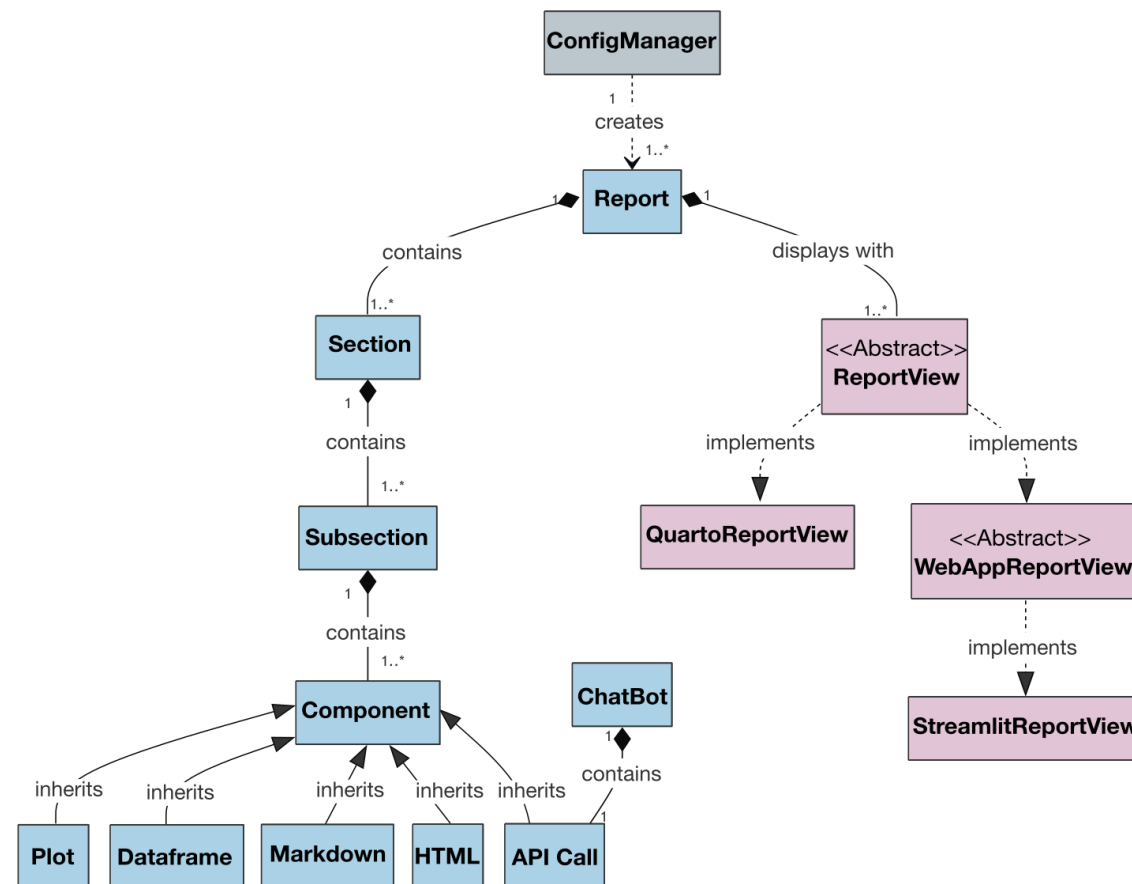
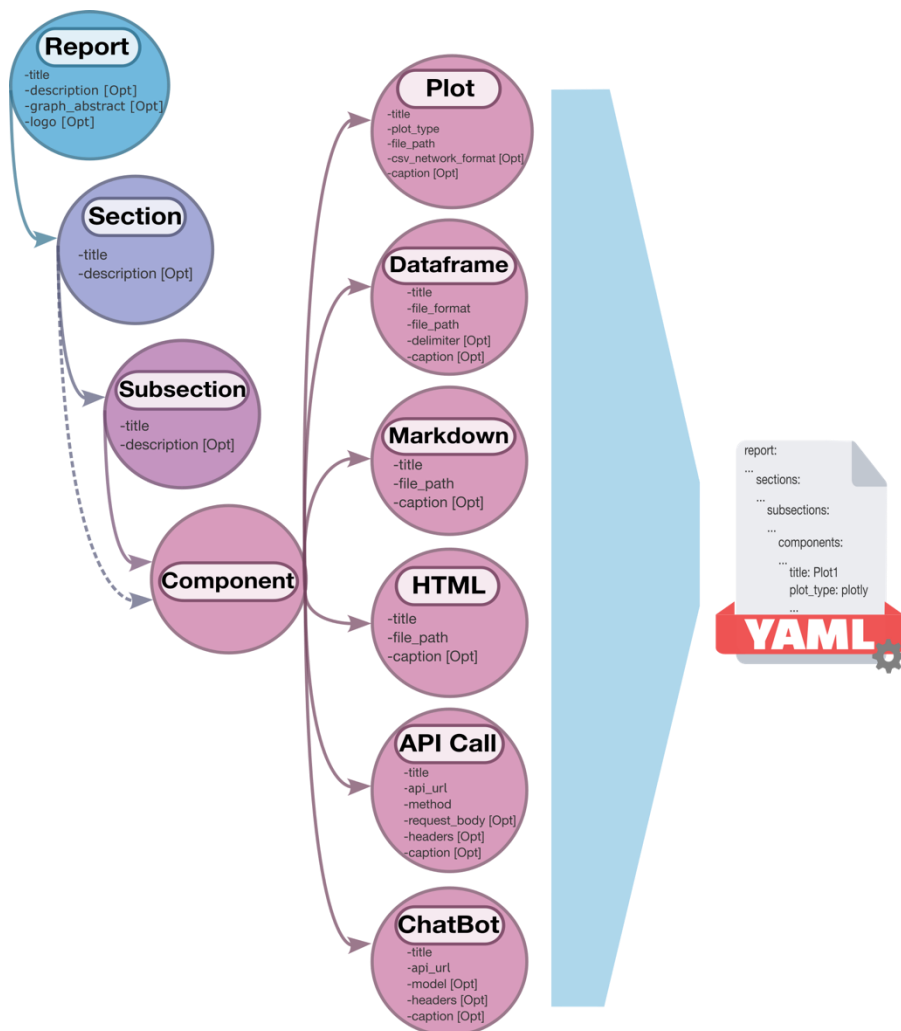
- Explore another dataset with the notebook:
- Try to run
`data/PXD041301/processed/PXD041301.sdrf_openms_design_msstats_in.csv`
- If you want to practice (by copy-pasting code), try to extend the other notebooks:
 - `3_data_analysis_PXD070899.ipynb`

Report Generation using Vuegen

- Turn results folder into different reports
- We collect plots and dataframes from our analysis example
- The analysis notebooks will export the relevant files



VueGen implementation details



See two examples

- Streamlit example: [link](#)
- Html-Report: [link](#)

Mass spectrometry-based
Proteomics introduction
documentation



Q Search * + K

QuantMS

QuantMS Hands-On for
PXD04621

QuantMS Input Files
Example datasets

Data Analysis

[Data Analysis Hands-On for
PXD04621](#)

Data Analysis PXD040621



vuegen -h

Create and start the streamlit app with the following command:

```
vuegen -dir data/PXD040621/report -st_autorun
```

Inspect streamlit report example online

A deployed version of the report can be accessed online at the following link:

- dsp-course-proteomics-intro-report.streamlit.app/

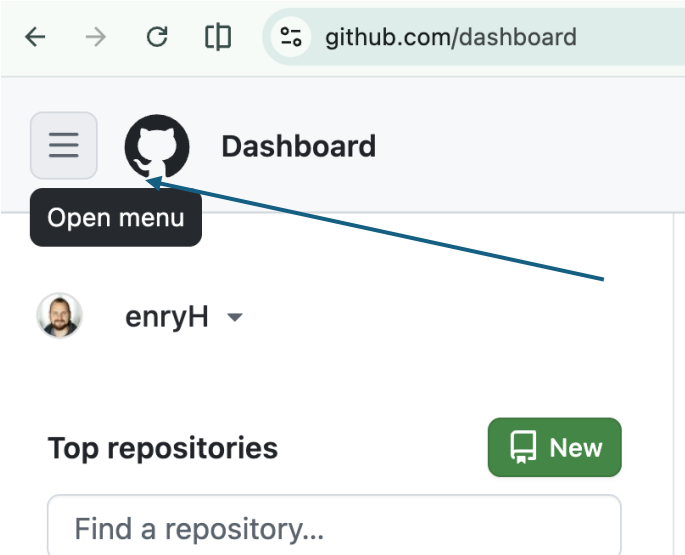
Download the VueGen HTML report

- [html report](#)

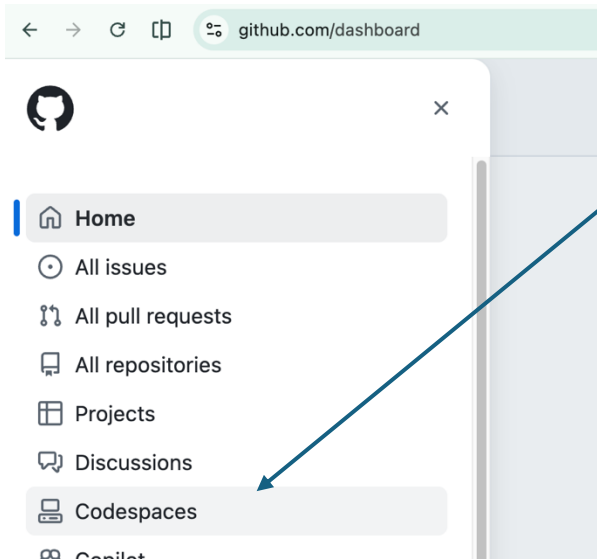
See instructions on [biosustain/dsp_course_proteomics_intro_report](https://biosustain.dsp_course_proteomics_intro_report).

Preparing a default analysis pipeline

- Tutorial can be basis for an automated analysis, see e.g https://github.com/biosustain/dsp_demo_nf_acore_vuegen
- Autogenerated VueGen Report is served on Seqera

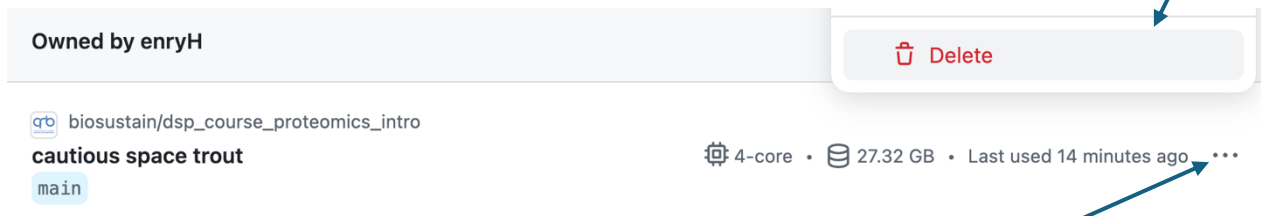


1.



2.

Delete your codespace after the course



4.

3.

45

Delete your codespace after the course

